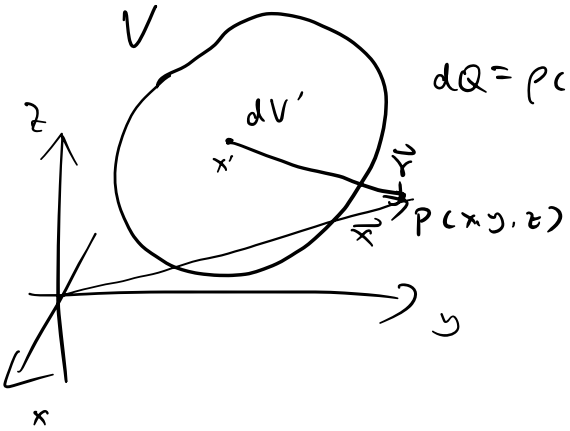
$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{QQ'}{r^3}\vec{r}$ ，引入场，$\vec{F} = Q'E$

$\Longrightarrow \vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^3}\vec{r}$

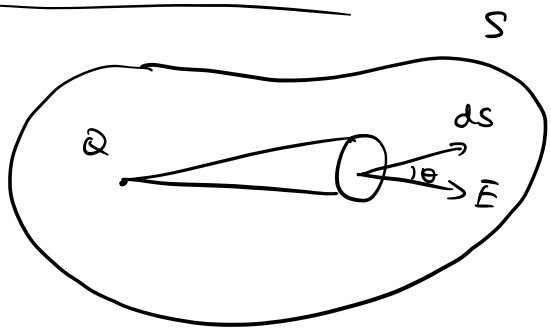
叠加：$\vec{E} = \sum_i \frac{1}{4\pi\varepsilon_0}\frac{Q_i}{r_i^3}\vec{r_i}$



$dQ = \rho(x')dV'$

$\vec{E}(\vec{x}) = \int_V \frac{\rho(x')dV'}{4\pi\varepsilon_0 r^3}\vec{r}$

<u>高斯定理 & 散度 div</u>



$\oint_S \vec{E}\cdot d\vec{s} = \frac{Q}{\varepsilon_0}$

Pf: $\vec{E}\cdot d\vec{s} = E\cos\theta\, ds = \frac{Q}{4\pi\varepsilon_0 r^2}\cos\theta\, ds$

$\Longrightarrow \oint_S \vec{E}\cdot d\vec{s} = \frac{Q}{4\pi\varepsilon_0}\oint d\Omega = \frac{Q}{\varepsilon_0}$

一般地，$\oint_S \vec{E}\cdot d\vec{s} = \frac{1}{\varepsilon_0}\sum_i Q_i$ （Q_i 在 S 内）

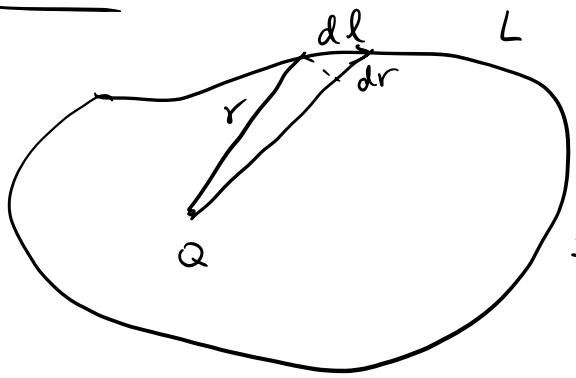
$\oint_S \vec{E}\cdot d\vec{s} = \frac{1}{\varepsilon_0}\int_V \rho dV$ （ρ 连续分布于空间中）

$\oint_S \vec{E}\cdot d\vec{s} = \int_V \mathrm{div}\vec{E}\, dV = \frac{1}{\varepsilon_0}\int_V \rho dV \Longrightarrow \mathrm{div}\vec{E} = \frac{\rho}{\varepsilon_0}$

$\Longrightarrow$ 电荷是电场的源

i.e. $\boxed{\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0}}$

旋度 rot



$$\oint_L \vec{E}\cdot d\vec{l} = \frac{Q}{4\pi\varepsilon_0}\oint \frac{\vec{r}}{r^3}\,d\vec{l}$$

$$\vec{r}\cdot d\vec{l} = r\cos\theta\, dl = r\,dr$$

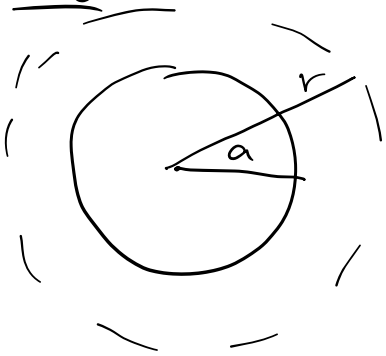
$$\Rightarrow \oint_L \vec{E}\cdot d\vec{l} = \frac{Q}{4\pi\varepsilon_0}\oint_L \frac{1}{r^2}dr = -\frac{Q}{4\pi\varepsilon_0}\oint_L d\left(\frac{1}{r}\right) = 0$$

$\Rightarrow$ 一个点电荷的电场环量为0

由叠加性，$\forall \vec{E}$，$\forall L$ 有 $\oint_L \vec{E}\cdot d\vec{l} = 0$

$$\oint_L \vec{E}\cdot d\vec{l} = \int_S (\mathrm{rot}\,\vec{E})\cdot d\vec{S} = 0 \Rightarrow \mathrm{rot}\,\vec{E} = 0 \quad \text{i.e. } \vec{\nabla}\times\vec{E} = 0$$

e.g. Q 均匀分布于半径 a 的球内，求各点 $\vec{E}$，计算 div $\vec{E}$



① $r>a$，$\oint_S \vec{E}\cdot d\vec{S} = 4\pi r^2 E = \frac{Q}{\varepsilon_0}$

$$\Rightarrow E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

$$\text{i.e. } \vec{E} = \frac{Q}{4\pi\varepsilon_0 r^3}\vec{r}$$

② $r<a$，$\frac{4}{3}\pi r^3\rho = \frac{4}{3}\pi r^3 \frac{Q}{\frac{4}{3}\pi a^3} = \frac{r^3}{a^3}Q$

$$\oint_S \vec{E}\cdot d\vec{S} = 4\pi r^2 E = \frac{r^3}{\varepsilon_0 a^3}Q \Rightarrow E = \frac{Qr}{4\pi\varepsilon_0 a^3}$$

$$\Rightarrow \vec{E} = \frac{Q}{4\pi\varepsilon_0 a^3}\vec{r}$$

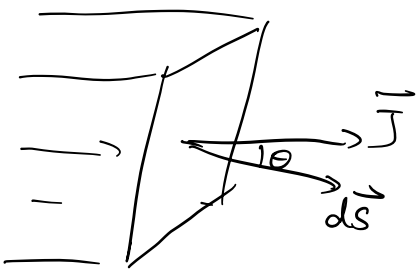
div：① $r>a$，$\vec{\nabla}\cdot\frac{\vec{r}}{r^3} = \frac{1}{r^2}(\vec{\nabla}\cdot\vec{e}_r) = 0$

$$\Rightarrow \vec{\nabla}\cdot\vec{E} = 0$$

② $r<a$，$\vec{\nabla}\cdot\vec{E} = \frac{Q}{4\pi\varepsilon_0 a^3}(\vec{\nabla}\cdot\vec{r}) = \frac{3Q}{4\pi\varepsilon_0 a^3} = \frac{\rho}{\varepsilon_0}$

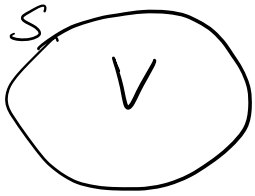
$\square$

电荷守恒定律



$$dI = J\,ds\cos\theta = \vec{J}\cdot d\vec{s}$$

$$I = \int_S \vec{J}\cdot d\vec{s} \qquad J = \sum_i \rho_i v_i$$



通过界面流出的总电流应等于V内电荷减少率

$$\oint_S \vec{J}\cdot d\vec{s} = -\int_V \frac{\partial\rho}{\partial t}\,dV$$

$$\| $$

$$\int_V \vec{\nabla}\cdot\vec{J}\,dV \quad \Longrightarrow \quad \vec{\nabla}\cdot\vec{J} + \frac{\partial\rho}{\partial t} = 0$$

电流连续性方程

* 若V是全空间，S为无穷远界面，有 $\frac{d}{dt}\int_V \rho\,dV = 0$ i.e. 全空间电荷守恒
* 恒定电流时，一切物理量不随t变化，$\frac{\partial\rho}{\partial t} = 0 \Rightarrow \vec{\nabla}\cdot\vec{J} = 0$ (恒定电流)

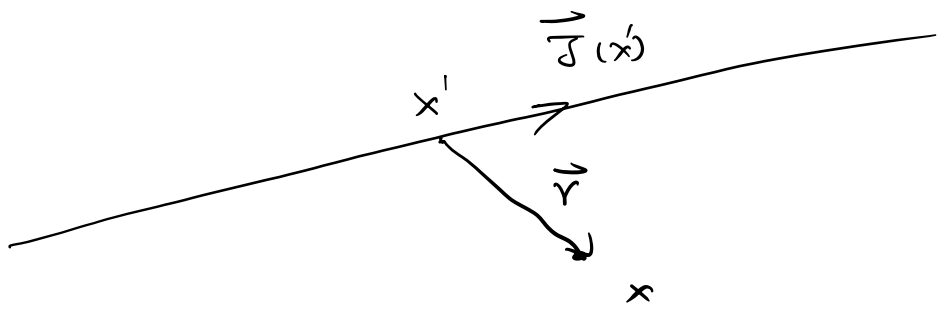
毕奥-萨伐尔定律

实验指出：一个电流元 $I\,d\vec{l}$ 在磁场中受力为

$$d\vec{F} = I\,d\vec{l}\times\vec{B}$$

Biot-Savart's law：设 $\vec{J}(x)$ 为源点 x' 的电流密度，r 为 x' 到场点 x 的矢径，则

$$B(x) = \frac{\mu_0}{4\pi}\int_V \frac{\vec{J}(x')\times\vec{r}}{r^3}\,dV' \qquad \mu_0\text{：真空磁导率}$$



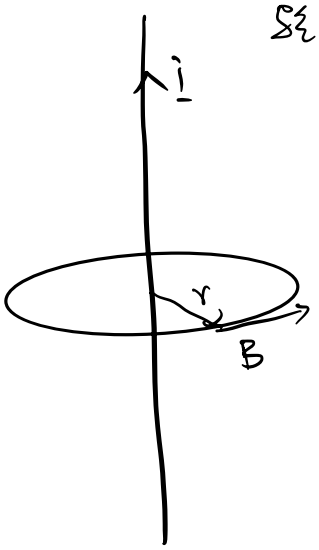
细导线：$\vec{J}\,dV' = \vec{J}\,dS_n\,dl = J\,dS_n\,d\vec{l}$

$$\Rightarrow \vec{B}(x) = \frac{\mu_0}{4\pi}\oint_L \frac{I\,d\vec{l}\times\vec{r}}{r^3}$$

磁场的环量和旋度

磁场沿闭合曲线的环量与通过闭合曲线所围曲面的电流成正比

$$\oint_L \vec{B}\cdot d\vec{l} = \mu_0 I \qquad \text{Ampère 环路定律}$$



验证：取无穷长直导线

$$\vec{B}(x) = \frac{\mu_0}{4\pi}\oint_L \frac{I\,d\vec{l}\times\vec{r}}{r^3} = \frac{\mu_0 I}{4\pi}\oint_L \frac{d\vec{l}\times\vec{r}}{r^3}$$

$$= \frac{\mu_0 I}{4\pi}\oint_L \frac{r\,dl\,\vec{e}_\varphi}{r^3}$$

$$= \frac{\mu_0 I}{4\pi}\vec{e}_\varphi\oint_L \frac{dl}{r^2} = \frac{\mu_0 I}{4\pi}\vec{e}_\varphi\oint_L \frac{r\,d\varphi}{\pi r^2}$$

$$= \frac{\mu_0 I}{4\pi}\vec{e}_\varphi\oint_L \frac{d\varphi}{\pi r} = \frac{\mu_0 I}{2\pi r}$$

$$\Rightarrow \oint_L \vec{B}\cdot d\vec{l} = \frac{\mu_0 I}{2\pi r}\cdot 2\pi r = \mu_0 I \qquad \square$$

$$\oint_L \vec{B}\cdot d\vec{l} = \mu_0\int_S \vec{J}\cdot d\vec{s}$$

把 L 缩小 $\Longrightarrow \vec{\nabla}\times\vec{B} = \mu_0\vec{J}$

磁场的散度

$\vec{B}$ 无源 $\Rightarrow \oint_S \vec{B}\cdot d\vec{S} = 0$ i.e. $\vec{\nabla}\cdot\vec{B} = 0$

Biot-Savart $\Longrightarrow \vec{\nabla}\times\vec{B} = \mu_0\vec{J}$, $\vec{\nabla}\cdot\vec{B} = 0$

$$\vec{B} = \frac{\mu_0}{4\pi}\int_V \frac{\vec{J}(x')\times\vec{r}}{r^3}\,dV = -\frac{\mu_0}{4\pi}\int_V \vec{J}(x')\times\vec{\nabla}\frac{1}{r}\,dV'$$

$$\vec{\nabla}\times\left(\vec{J}(x')\frac{1}{r}\right) = \left(\nabla\frac{1}{r}\right)\times\vec{J}(x') + \underbrace{\frac{1}{r}\,\vec{\nabla}\times\vec{J}(x')}_{=0}$$

($\vec{\nabla}$ acts on x', not x)

$$\Rightarrow \vec{B} = \frac{\mu_0}{4\pi}\int_V \vec{\nabla}\times\left(\vec{J}(x')\frac{1}{r}\right)dV'$$

$$= \frac{\mu_0}{4\pi}\vec{\nabla}\times\int_V \frac{\vec{J}(x')}{r}\,dV' = \vec{\nabla}\times\vec{A}$$

$$\left(\vec{A} = \frac{\mu_0}{4\pi}\int_V \frac{\vec{J}(x')\,dV'}{r}\right)$$

$$\Rightarrow \boxed{\vec{\nabla}\cdot\vec{B} = \vec{\nabla}\cdot(\vec{\nabla}\times\vec{A}) = 0}$$

$$\vec{\nabla}\times\vec{B} = \vec{\nabla}\times(\vec{\nabla}\times\vec{A}) = \vec{\nabla}(\vec{\nabla}\cdot\vec{A}) - \vec{\nabla}^2\vec{A}$$

$$\vec{\nabla}\cdot\vec{A} = \frac{\mu_0}{4\pi}\int_V \vec{\nabla}\cdot\frac{\vec{J}(x')}{r}\,dV = \frac{\mu_0}{4\pi}\int_V \vec{J}(x')\cdot\vec{\nabla}\frac{1}{r}\,dV'$$

因为 $r = |x - x'| \Longrightarrow$

$$\vec{\nabla}\cdot\vec{A} = -\frac{\mu_0}{4\pi}\int_V \vec{J}(x')\cdot\vec{\nabla}'\frac{1}{r}\,dV'$$

$$= -\frac{\mu_0}{4\pi}\int_V \nabla'\cdot\left(\vec{J}(x')\frac{1}{r}\right)dV + \frac{\mu_0}{4\pi}\int_V \frac{1}{r}\vec{\nabla}'\cdot\vec{J}(x')\,dV'$$

$$= 0 + \frac{\mu_0}{4\pi}\int_V \frac{1}{r}\underbrace{\vec{\nabla}'\cdot\vec{J}(x')}_{\text{恒定电流}}\,dV' = 0$$

$$\vec{\nabla}^2\vec{A} = \frac{\mu_0}{4\pi}\int_V \vec{J}(x')\,\vec{\nabla}^2\frac{1}{r}\,dV'$$

$$= -\frac{\mu_0}{4\pi}\int_V \vec{J}(x')\,\vec{\nabla}\cdot\frac{\vec{r}}{r^3}\,dV'$$

$r\neq 0$ 时，$\vec{\nabla}\cdot\frac{\vec{r}}{r^3} = \left(\vec{\nabla}\frac{1}{r^3}\right)\cdot\vec{r} + \frac{1}{r^3}\vec{\nabla}\cdot\vec{r}$

$$= -3\frac{1}{r^4}\vec{\nabla}r\cdot\vec{r} + \frac{1}{r^3}\left(\frac{\partial(x-x')}{\partial x} + \frac{\partial(y-y')}{\partial y} + \frac{\partial(z-z')}{\partial z}\right)$$

$$= -3\frac{1}{r^4}\frac{\vec{r}}{r}\cdot\vec{r} + \frac{3}{r^3}$$

$$= -3\frac{1}{r^3} + \frac{3}{r^3} = 0$$

$\Rightarrow$ 被积函数在 $x'=x$ 不为零，因此体积分只需对包围 x 点的小球积分，取 $\vec{J}(x') = \vec{J}(x)$

$$\int_V \vec{\nabla}\cdot\frac{\vec{r}}{r^3}\,dV' = -\int_V \vec{\nabla}'\cdot\frac{\vec{r}}{r^3}\,dV' = -\oint_S \frac{\vec{r}}{r^3}\cdot d\vec{S}'$$

$$= \oint_S \frac{1}{r^2}\,dS = \oint d\Omega = 4\pi$$

$$\Rightarrow \vec{\nabla}^2\vec{A} = -\mu_0\vec{J}$$

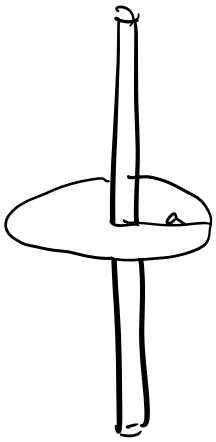
$$\Rightarrow \vec{\nabla}\times\vec{B} = \mu_0\vec{J}$$

* Biot-Savart's law $\Longrightarrow$ $\vec{\nabla}\cdot\vec{B} = 0$ 一般也成立

恒定电流成立

$\Longrightarrow$ $\vec{\nabla}\times\vec{B} = \mu_0\vec{J}$ 恒定电流成立

e.g. 均匀分布在半径 a 的无穷长直导线，求空间各点 $\vec{B}$，$\vec{B}$ 旋度.



$r>a$ 时，$\oint_L \vec{B}\cdot d\vec{l} = 2\pi r B = \mu_0 I$

$$\Rightarrow B = \frac{\mu_0 I}{2\pi r} \qquad \vec{B} = \frac{\mu_0 I}{2\pi r}\vec{e}_\phi \quad (r>a)$$

$r<a$ 时，$\pi r^2 J = \pi r^2 \frac{I}{\pi a^2} = \frac{r^2}{a^2} I$

$$\oint_L \vec{B}\cdot d\vec{l} = 2\pi r B = \mu_0 \frac{r^2}{a^2} I$$

$$\Rightarrow B = \frac{\mu_0 r I}{2\pi a^2}, \qquad \vec{B} = \frac{\mu_0 r I}{2\pi a^2}\vec{e}_\phi \quad (r<a)$$

$r>a$ 时，
$$\vec{\nabla}\times\vec{B} = \left(\frac{1}{r}\frac{\partial B_z}{\partial \phi} - \frac{\partial B_\phi}{\partial z}\right)\vec{e}_r + \left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r}\right)\vec{e}_\phi + \left(\frac{1}{r}\frac{\partial}{\partial r}(r B_\phi) - \frac{1}{r}\frac{\partial B_r}{\partial \phi}\right)\vec{e}_z$$

$$= -\frac{B_\phi}{\partial z}\vec{e}_r + \frac{1}{r}\frac{\partial}{\partial r}(r B_\phi)\vec{e}_z = 0$$

$r<a$ 时，
$$\vec{\nabla}\times\vec{B} = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\mu_0 r I}{2\pi a^2}\right)\vec{e}_z$$

$$= \frac{1}{r}\frac{\mu_0 I}{2\pi a^2}\cdot 2r\,\vec{e}_z$$

$$= \frac{\mu_0 I}{\pi a^2}\vec{e}_z = \mu_0 \vec{J}$$

$\square$

电磁感应定律 Faraday

闭合线圈中的感应电动势与通过该线圈内部的磁通量变化成反比

$$\mathcal{E} = -\frac{d}{dt}\int_S \vec{B}\cdot d\vec{S}$$

$$\oint_L \vec{E}\cdot d\vec{l} = -\frac{d}{dt}\int_S \vec{B}\cdot d\vec{S}$$

固定 L，$\oint_L \vec{E}\cdot d\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t}\cdot d\vec{S}$

$$\Longrightarrow \vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

位移电流

非恒定时有 $\vec{\nabla}\cdot\vec{J} = -\frac{\partial \rho}{\partial t}$

对于 $\vec{\nabla}\times\vec{B} = \mu_0\vec{J}$，由于 $\vec{\nabla}\cdot\vec{\nabla}\times\vec{B} = 0 \Longrightarrow \vec{\nabla}\cdot\vec{J} = 0$ 矛盾

假设存在一个位移电流 J_D

$$\vec{\nabla}\cdot(\vec{J}+\vec{J}_D) = 0 \rightsquigarrow \text{数学上不合理}$$

$$\vec{\nabla}\times\vec{B} = \mu_0(\vec{J}+\vec{J}_D)$$

由 $\begin{cases}\vec{\nabla}\cdot\vec{J} + \frac{\partial \rho}{\partial t} = 0 \\ \vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0}\end{cases} \Longrightarrow \vec{\nabla}\cdot\left(\vec{J} + \varepsilon_0\frac{\partial \vec{E}}{\partial t}\right) = 0$

$$\Longrightarrow \vec{J}_D = \varepsilon_0\frac{\partial \vec{E}}{\partial t}$$

Maxwell eqs

$$\begin{cases} \vec{\nabla} \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \dfrac{\partial \vec{E}}{\partial t} \\ \vec{\nabla} \cdot \vec{E} = \dfrac{\rho}{\varepsilon_0} \\ \vec{\nabla} \cdot \vec{B} = 0 \end{cases}$$

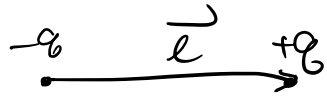
洛仑兹力

- $q \longrightarrow \vec{v}$ $\qquad \vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$

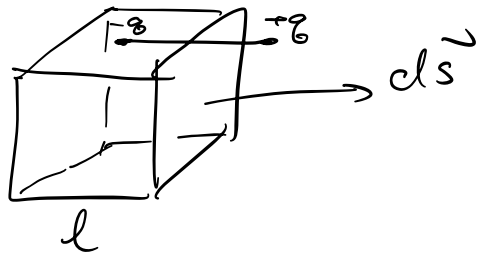
介质极化

宏观电偶极矩分布用电极化强度矢量 $\vec{P}$ 描述，它等于小体积 ΔV 内的总电偶极矩与 ΔV 之比

$$\vec{P} = \frac{\sum_i \vec{p}_i}{\Delta V}$$



$$\vec{p} = q\vec{l}$$



设单位体积分子数为 n，则穿出 $d\vec{s}$ 外的正电荷为

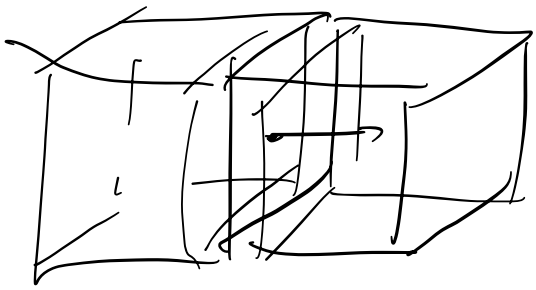
$$nq\vec{l}\cdot d\vec{s} = n\vec{p}\cdot d\vec{s} = \vec{P}\cdot d\vec{s}$$

由 V 内穿出的正电荷为 $\oint_S \vec{P}\cdot d\vec{s}$

也等于 V 内束缚电荷 ρ：束缚

$$\int_V \rho_p \, dV = -\oint_S \vec{P}\cdot d\vec{s}$$

$$\Rightarrow \rho_p = -\vec{\nabla}\cdot\vec{P}$$



$$\sigma_p ds = -(\vec{P}_2 - \vec{P}_1)\cdot d\vec{s}$$

$$\Rightarrow \sigma_p = -\vec{e}_n\cdot(\vec{P}_2 - \vec{P}_1)$$

若在 Maxwell's eqs 中电荷密度 ρ 包括自由电荷密度 ρ_f 和束缚电荷密度 ρ_p，则在介质内 $\varepsilon_0 \vec{\nabla}\cdot\vec{E} = \rho_f + \rho_p$ 仍成立

$$\vec{\nabla}\cdot(\varepsilon_0\vec{E} + \vec{P}) = \rho_f$$

引入电位移矢量 $\vec{D} = \varepsilon_0\vec{E} + \vec{P}$

$$\Rightarrow \vec{\nabla}\cdot\vec{D} = \rho_f$$

$$\vec{P} = \underbrace{\chi_e}_{\text{极化率}}\varepsilon_0\vec{E} \quad \Rightarrow \vec{D} = \varepsilon_0\vec{E} + \chi_e\varepsilon_0\vec{E}$$

$$= (\varepsilon_0 + \chi_e\varepsilon_0)\vec{E}$$

$$= \underbrace{\varepsilon_0\underbrace{(1+\chi_e)}_{\text{相对电容率}}}_{\text{电容率}}\vec{E}$$